metal ions on the surface of the cells and reducing ions in the presence of enzymes.

The biosynthesized nanoparticles have been used in a variety of applications including drug carriers for targeted delivery, cancer treatment, gene therapy and DNA analysis, antibacterial agents, biosensors, enhancing reaction rates, separation science, and magnetic resonance imaging (MRI). Many microorganisms can produce inorganic nanoparticles through either intracellular or extracellular routes. This section describes the production of various nanoparticles via biological methods following the categories of metallic nanoparticles including gold, silver, alloy and other metal nanoparticles, oxide nanoparticles consisting of magnetic and nonmagnetic oxide nanoparticles, sulfide nanoparticles, and other miscellaneous nanoparticles (Figure 1.6).

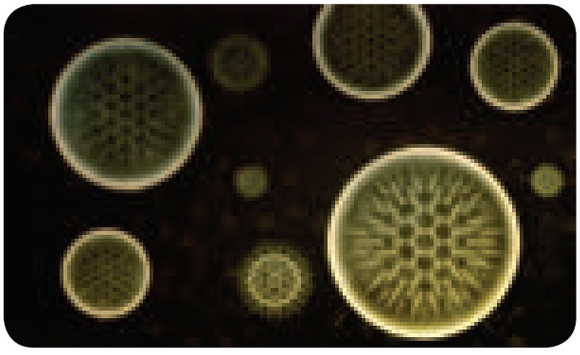


Figure 1.6: Advanced Nano particles

1.6 Equipments

1.6.1 Confocal Microscopy

Confocal microscopy offers several advantages over conventional optical microscopy, including shallow depth of field, elimination of out-of-focus glare, and the ability to collect serial optical sections from thick specimens. In the biomedical sciences, a major application of confocal microscopy involves imaging either fixed or living cells and tissues that have usually been labeled with one or more fluorescent probes.

Confocal microscopy, most frequently **confocal laser scanning microscopy (CLSM)** or **laser confocal scanning microscopy (LCSM)**, is an optical imaging technique for increasing optical resolution and contrast of a micrograph by means of using a spatial pinhole to block out-of-focus light in image formation. Capturing multiple two-dimensional images at different depths in a sample enables the reconstruction of three-dimensional structures (a process known as optical sectioning) within an object. This technique is used extensively in the scientific and industrial communities and typical applications are in life sciences, semiconductor inspection and materials science.

Light travels through the sample under a conventional microscope as far into the specimen as it can penetrate, while a confocal microscope only focuses a smaller beam of light at one narrow depth level at a time. The CLSM achieves a controlled and highly limited depth of focus.

1.6.2 DNA Sequencing System

Sequencing means finding the order of nucleotides on a piece of DNA. Nucleotide order determines amino acid order, and by extension, protein structure and function (proteomics). An alteration in a DNA sequence can lead to an altered or non functional protein, and hence to a genetic disorder. DNA sequence is important to detect the type of mutations in genetic diseases and offer hope for the eventual development of treatment DNA.

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